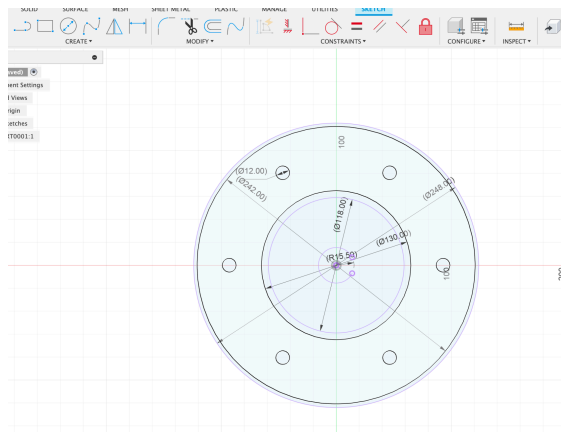


EXPERIMENT AIM:

To prepare the flange assembly using Fusion 360

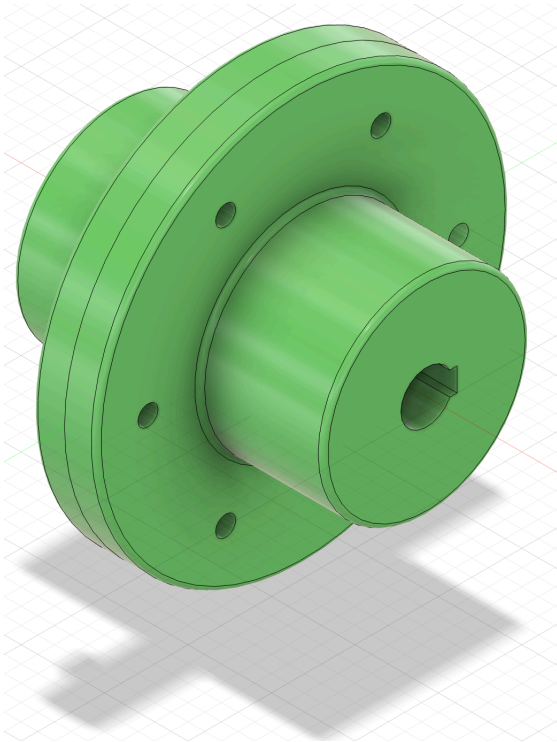
1. Creating Flange

- Sketch $\varnothing 190$ mm circle → Extrude 20 mmx

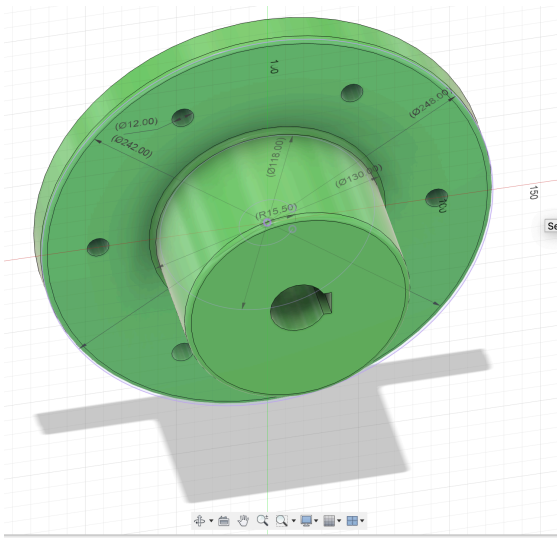


- Sketch $\varnothing 90$ mm → Extrude 55 mm
- Create $\varnothing 12$ mm holes at 70 mm radius
- Use Circular Pattern (4 holes)

- Apply Extruded Cut → Through All



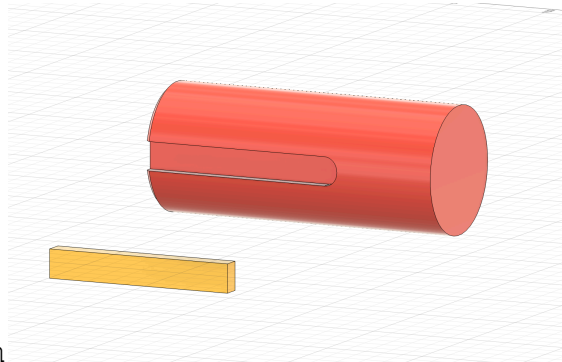
- Create rectangular slot 14 × 4 mm → cut through



2. Creating End Shaft

- Sketch Ø45 mm circle → Extrude 100 mm
- Sketch rectangle slot 14 mm width

- Position: 17.5 mm from center



- Extruded Cut → 90 mm
- Apply Fillet (5 mm)

3. Creating Taper Pin

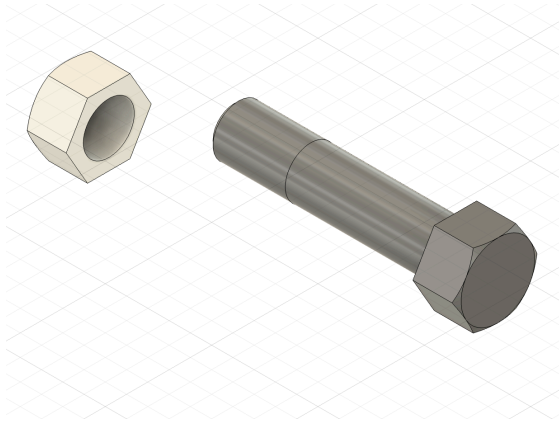
- Sketch 14 × 7 mm rectangle
- Extrude 80 mm
- Apply Draft = $\arctan(1/100)$

4. Creating Bolt & Nut

Bolt:

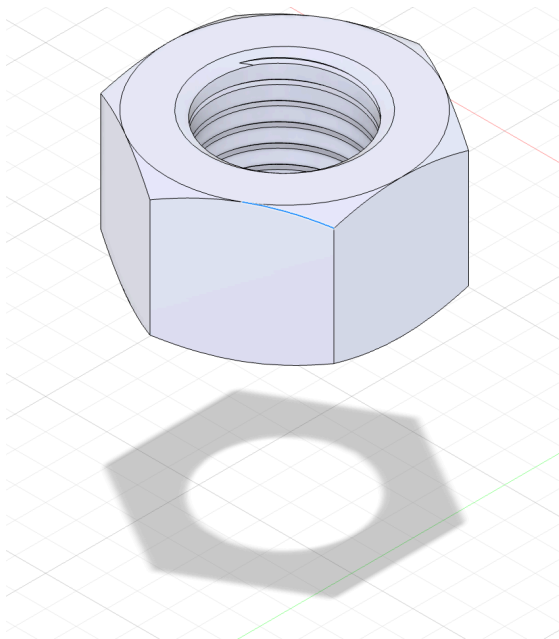
- Sketch Ø12 mm → Extrude 52 mm

- Create hex head → Extrude 8 mm



Nut:

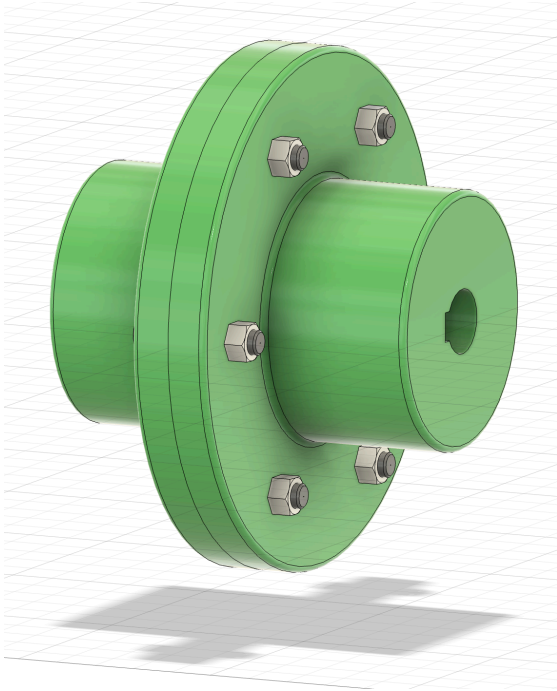
- Sketch hex + $\varnothing 12$ mm hole
- Extrude 10 mm
- Apply Thread (M12 $\times$ 1.5)



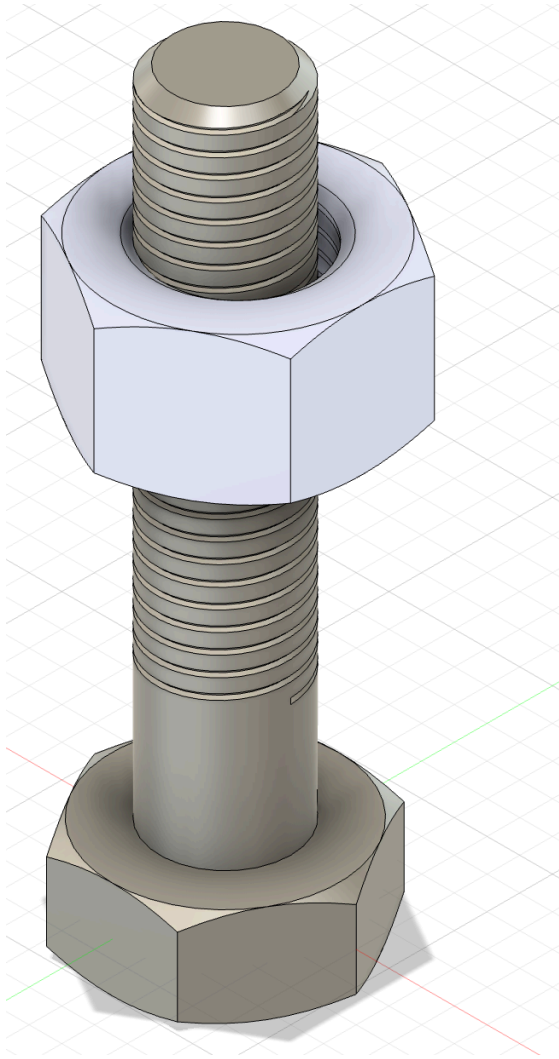
5. Assembly

- Insert: 2 Flanges, Shafts, Bolts, Nuts, Pins

- Apply mates:
- Flange faces → coincident



- Shaft & flange → concentric



- Bolt & nut → aligned

- Pins → fixed in slots.

